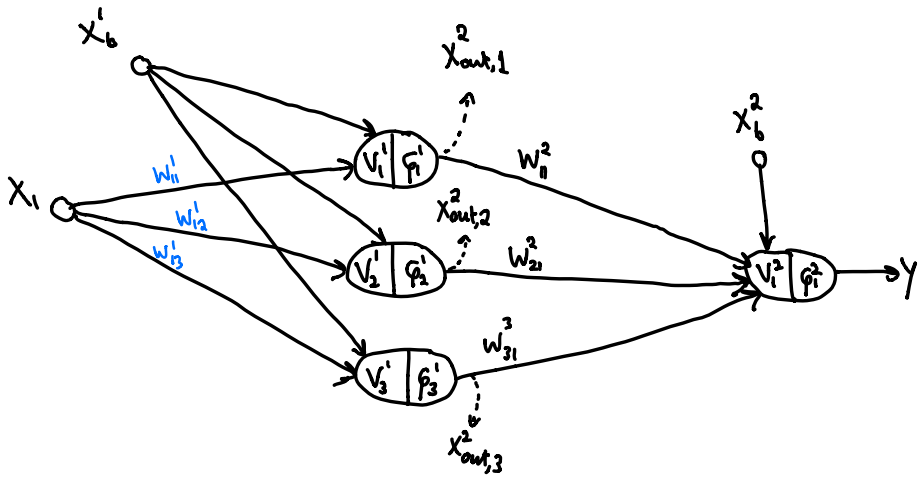
Task 1: Regression



$$V_1^1 = W_{11}^1 X_1 + X_b^1$$
$$V_2^1 = W_{12}^1 X_1 + X_b^1$$
$$V_3^1 = W_{13}^1 X_1 + X_b^1$$

OR

$$V_1^1 = W_{11}^1 X_1 + W_{21}^1 X_b^1$$
$$V_2^1 = W_{12}^1 X_1 + W_{22}^1 X_b^1$$
$$V_3^1 = W_{13}^1 X_1 + W_{23}^1 X_b^1$$

$$X_{out,1}^2 = \varphi_1^1(V_1^1)$$
$$X_{out,2}^2 = \varphi_2^1(V_2^1)$$
$$X_{out,3}^2 = \varphi_3^1(V_3^1)$$

$$y_1 = W_{11}^2 X_{out,1}^2 + W_{21}^2 X_{out,2}^2 + W_{31}^2 X_{out,3}^3 + X_b^2 \Longleftrightarrow$$

$$\Longleftrightarrow y_1 = W_{11}^2 \varphi_1^1(V_1^1) + W_{21}^2 \varphi_2^1(V_2^1) + W_{31}^2 \varphi_3^1(V_3^1) + X_b^2$$

$W^1_{11} = 0.5$, $W^1_{12} = -0.3$, $W^1_{13} = 0.8$

$W^1_{21} = 0.1$, $W^1_{22} = -0.2$, $W^1_{23} = 0.3$

$W^2_{11} = 0.7$, $W^2_{21} = -0.5$, $W^2_{31} = 0.4$

$X^1_b = 0.05$, $X^2_b = 0.03$

OR

$W^1_{11} = 0.5$, $W^1_{12} = -0.3$, $W^1_{13} = 0.8$

$W^1_{21} = 0.1$, $W^1_{22} = -0.2$, $W^1_{23} = 0.3$

$W^2_{11} = 0.7$, $W^2_{21} = -0.5$, $W^2_{31} = 0.4$

For $x_1 = -1$:

$V^1_1 = 0.5 \cdot (-1) + 0.05 = -0.45$

$V^1_2 = -0.3(-1) + 0.05 = 0.35$

$V^1_3 = 0.8(-1) + 0.05 = -0.75$

OR

$V^1_1 = 0.5 \cdot (-1) + 0.1 \cdot 0.05 = -0.495$

$V^1_2 = -0.3(-1) - 0.2 \cdot 0.05 = 0.29$

$V^1_3 = 0.8(-1) + 0.3 \cdot 0.05 = -0.785$

$$\varphi(V) = \tanh = \frac{e^{v} - e^{-v}}{e^{v} + e^{-v}}:$$

$$\varphi^1_1(V^1_1) = \frac{e^{-0.45} - e^{0.45}}{e^{-0.45} + e^{0.45}} = -0.42190$$

$$\varphi^1_2(V^1_2) = \frac{e^{0.35} - e^{-0.35}}{e^{0.35} + e^{-0.35}} = 0.33638$$

$$\varphi^1_3(V^1_3) = \frac{e^{-0.75} - e^{0.75}}{e^{-0.75} + e^{0.75}} = -0.63515$$

OR

$$\varphi^1_1(V^1_1) = \frac{e^{-0.495} - e^{0.495}}{e^{-0.495} + e^{0.495}} = -0.45818$$

$$\varphi^1_2(V^1_2) = \frac{e^{0.29} - e^{-0.29}}{e^{0.29} + e^{-0.29}} = 0.28213$$

$$\varphi^1_3(V^1_3) = \frac{e^{-0.785} - e^{0.785}}{e^{-0.785} + e^{0.785}} = -0.65557$$

$y_1 = 0.7 \cdot (-0.42190) - 0.5 \cdot 0.33638 + 0.4 \cdot (-0.63515) + 0.03$

OR $y_1 = 0.7 \cdot (-0.45818) - 0.5 \cdot 0.28213 + 0.4 \cdot 0.28213 + 0.03$

$d_1 = -1$ $y_1 = -0.68758$

OR $y_1 = -0.31894$

Last layer:

$$L = \frac{1}{2}\left(d_1 - y_1\right)^2 = \frac{1}{2}\left(d_1 - \varphi_1^2\left(v_1^2\right)\right)^2 = \frac{1}{2}\left(-1 + 0.68758\right)^2$$

$$v_1^2 = w_{11}^2 x_{out,1}^2 + w_{21}^2 x_{out,2}^2 + w_{31}^2 x_{out,3}^3 + x_b^2$$

$$w_{11,new}^2 = w_{11,old}^2 - \eta \frac{\partial L}{\partial w_{11}^2} = w_{11,old}^2 - \eta \frac{\partial L}{\partial v_1^2} \frac{\partial v_1^2}{\partial w_{11}^2}$$

$$\frac{\partial v_1^2}{\partial w_{11}^2} = x_{out,1}^2$$

not tanh activation function, its a linear function.

$\therefore \varphi(v) = v \therefore \frac{\partial \varphi}{\partial v} = 1$

$$\frac{\partial L}{\partial v_1^2} = \left(d_1 - \varphi_1^2\left(v_1^2\right)\right) \frac{\partial \varphi}{\partial v_1^2} = \left(d_1 - \varphi_1^2\left(v_1^2\right)\right) \cdot 1$$

$$w_{11,new}^2 = w_{11,old}^2 + \eta \left(d_1 - \varphi_1^2\left(v_1^2\right)\right) x_{out,1}^2$$

$$\left(1 - \left[\varphi_1^2\left(v_1^2\right)\right]^2\right) = -\delta_1^2$$

1st Layer:

$$x_{out,1}^2 = \varphi_1^1\left(v_1^1\right)$$

$$v_1^1 = w_{11}^1 x_1 + x_b^1$$

w_{11}^1 in v_1^1

V_1^1 in $X_{out,1}^2$

$X_{out,1}^2$ in V_1^2

V_1^2 in L

$$\frac{\partial L}{\partial w_{11}^1} = \frac{\partial L}{\partial v_1^2} \frac{\partial v_1^2}{\partial x_{out,1}^2} \frac{\partial x_{out,1}^2}{\partial v_1^1} \frac{\partial v_1^1}{\partial w_{11}^1}$$

$$\frac{\partial L}{\partial v_1^2} = \left(d_1 - \varphi_1^2(v_1^2)\right)$$

$$\frac{\partial v_1^2}{\partial x_{out,1}^2} = w_{11}^2$$

$$\frac{\partial x_{out,1}^2}{\partial v_1^1} = \frac{\partial \varphi_1^1(v_1^1)}{\partial v_1^1} = 1 - \left[\varphi_1^1(v_1^1)\right]^2$$

$$\frac{\partial v_1^1}{\partial w_{11}^1} = x_1$$

$$\therefore \frac{\partial L}{\partial w_{11}^1} = \left(d_1 - \varphi_1^2(v_1^2)\right) \cdot w_{11}^2 \left(1 - \left[\varphi_1^1(v_1^1)\right]^2\right) \cdot x_1$$

$$w_{11,new}^1 = w_{11,old}^1 - \eta \frac{\partial L}{\partial w_{11}^1} = w_{11,old}^1 - \eta \left(d_1 - \varphi_1^2(v_1^2)\right) \cdot w_{11}^2 \left(1 - \left[\varphi_1^1(v_1^1)\right]^2\right) \cdot x_1$$